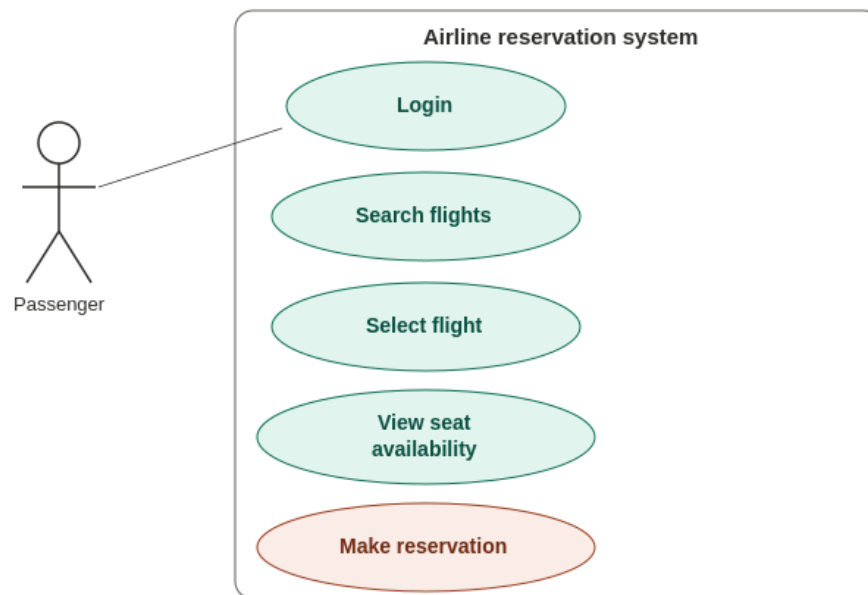


Use Case Diagram for Airline system

The diagram below shows the passenger actor and the five core use cases supported by the system: logging in, searching flights, selecting a flight, viewing seat availability, and making a reservation.

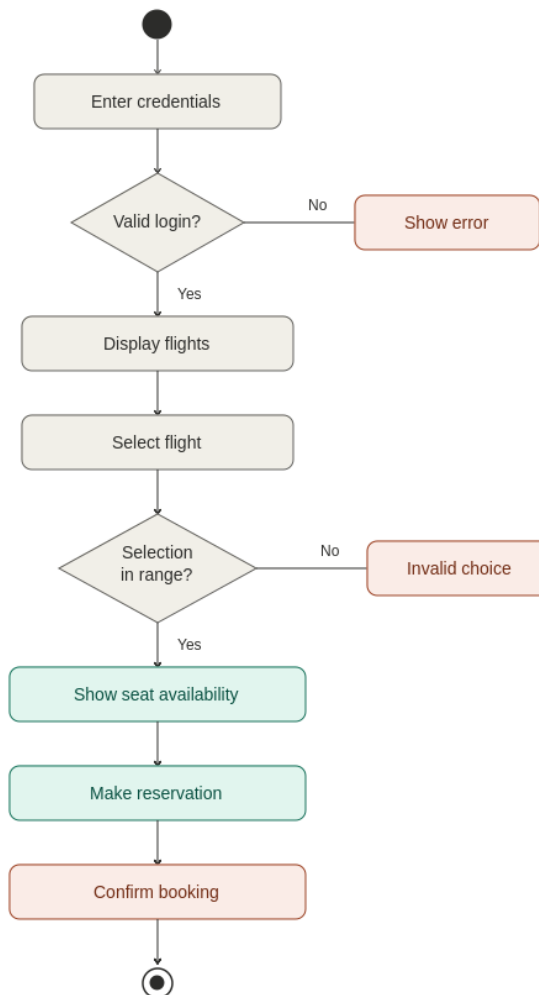


Use case summary

Use case	Actor	Description
Login	Passenger	Passenger enters a username and password; the system authenticates the credentials.
Search flights	Passenger	System displays the list of available flights to the authenticated passenger.
Select flight	Passenger	Passenger chooses a flight from the displayed list by entering its number.
View seat availability	Passenger	System shows which seats are open on the selected flight.
Make reservation	Passenger	Passenger books a seat on the selected flight and receives a confirmation.

Activity Diagram for Airline system

The activity diagram traces the control flow through the reservation process: credential entry and validation, flight listing and selection with range validation, seat availability, and final booking confirmation. Invalid input at either decision point routes to an error message instead of continuing the flow.



Key decision points

- Valid login?
 - checks the result of `authenticateUser()`; an invalid login shows an error and does not proceed to the flight list.
- Selection in range?
 - checks that the chosen flight number falls between 1 and 3; an out-of-range choice shows "Invalid choice" instead of continuing to seat availability.